ModelIt!

Teacher's Guide

Can We Stop an Earthquake from Breaking Buildings?

Engineering Solutions to Protect People from Natural Hazards

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-ESS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-ESS3-2

Generate and compare multiple solutions to reduce the impacts of natural Earth processes on humans.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students generate and compare multiple engineering solutions to reduce earthquake damage, using model evidence to evaluate their effectiveness.
Disciplinary Core Idea	**ESS3.B Natural Hazards** — A variety of hazards result from natural processes. Humans cannot eliminate the hazards but can take steps to reduce their impacts.
Crosscutting Concept	**Cause and Effect** — Students identify cause and effect relationships between earthquake strength, building design, and damage levels to engineer protective solutions.

CAST Testing Alignment

MULTIPLE CHOICE:

Two buildings experience the same earthquake. Building A is made of stiff concrete with no flexibility. Building B has a steel frame with cross-bracing that allows it to sway. Which building is more likely to survive, and why?

CONSTRUCTED RESPONSE:

Describe two different engineering solutions that help buildings survive earthquakes. Explain how each one works to reduce damage.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Earthquake engineering is one of the most important applications of science to human safety. Engineers cannot prevent earthquakes, but they have developed remarkable technologies that allow buildings to survive even the most powerful shaking.

How Earthquakes Damage Buildings

Earthquakes damage buildings through ground shaking, which causes structures to vibrate. If the shaking frequency matches the building's natural frequency (resonance), the vibrations amplify dramatically. Stiff, rigid buildings tend to crack and crumble because they resist the movement. The most dangerous situation is when heavy concrete buildings on soft soil experience a strong earthquake — the soft soil amplifies the shaking.

Flexible Design Philosophy

Modern earthquake engineering follows the principle that buildings should bend, not break. Steel-frame buildings with cross-bracing can sway several feet during an earthquake and return to their original position. This flexibility absorbs the earthquake's energy gradually rather than resisting it rigidly. Think of it like a palm tree in a hurricane — it bends with the wind and survives, while a stiff oak might snap.

Base Isolation Technology

Base isolation is one of the most effective earthquake protection technologies. The building sits on special rubber and steel pads (isolators) that allow the foundation to move with the ground while the building above stays relatively still. During an earthquake, the ground may move several inches, but the building barely moves. This technology is used in hospitals, museums, and critical infrastructure worldwide.

Real-World Success Stories

Japan's Yokohama Landmark Tower (296m tall) uses a combination of base isolation and active dampers to survive earthquakes. During the 2011 magnitude 9.0 earthquake, buildings designed with modern seismic technology performed remarkably well while older buildings suffered severe damage. Chile's building codes, updated after the 1960 magnitude 9.5 earthquake (the largest ever recorded), have saved countless lives in subsequent earthquakes.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify earthquake strength, building flexibility, and damage level as the key components of the earthquake engineering system.

E **Establish Relationships**
Students determine that earthquake strength positively affects damage level (stronger quake = more damage), while building flexibility negatively affects damage level (more flexibility = less damage).

V **Visualize & Model**
Students build a model showing how earthquake intensity interacts with building design to determine how much damage occurs.

E **Evaluate Outcomes**
Students run scenarios comparing stiff vs flexible buildings in the same earthquake to see the dramatic difference in damage levels.

R **Revise & Extend**
Students add foundation depth or ground type to explore more complex factors in earthquake engineering.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Imagine the ground starts shaking right now. Would this building survive? Let us find out!"*

Watch For

Show a quick video of earthquake shaking. Ask: What would you do if this happened?

Slide 2: Learning Objectives

2 min

Say: *"Today we become earthquake engineers — people who design buildings to survive nature's most powerful force."*

Watch For

Have students read objectives aloud. Pre-teach flexible and base isolation.

Slide 3: Big Question

3 min

Say: *"Some buildings crumble in an earthquake while the building right next door stays standing. WHY?"*

Watch For

Show before-and-after photos of earthquake damage. Ask students why some buildings survived.

Slide 4: LEVER Framework

2 min

Say: *"We will build a model to discover the secret of earthquake-proof design."*

Watch For

Preview LEVER steps. Connect to the engineering design process.

Slide 5: Activity 1: Components

7 min

Say: *"What determines whether a building survives or collapses?"*

Watch For

Guide component sorting. Discuss why earthquake strength is external (we cannot control it).

Slide 6: Activity 2: Connections

7 min

Say: *"Stronger earthquake means MORE damage — positive. But more flexibility means LESS damage — negative! Engineers use that negative relationship to save lives."*

Watch For

Help students draw arrows. Emphasize the critical negative relationship.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us hit a stiff building and a flexible building with the same earthquake and see what happens!"*

Watch For

Students predict first. Run both scenarios side by side for dramatic comparison.

Slide 8: Discoveries

4 min

Say: *"Engineers cannot stop earthquakes — but they CAN stop buildings from falling down!"*

Watch For

Lead discussion about how flexibility saves buildings. Connect to real engineering examples.

Slide 9: STEM Challenge

8 min

Say: *"Now YOU are the engineers. Build a structure that survives the shake table!"*

Watch For

Distribute materials (craft sticks, marshmallows, tape). Groups build and test structures.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Earthquake Strength
- **INTERNAL:** Building Flexibility, Damage Level

Relationships

- **Earthquake Strength to Damage Level:** POSITIVE (+) — Stronger earthquakes shake the ground more violently, causing more damage to buildings. A magnitude 7 earthquake is about 30 times more powerful than a magnitude 6.
- **Building Flexibility to Damage Level:** NEGATIVE (-) — More flexible buildings absorb earthquake energy by swaying instead of cracking. Increasing flexibility DECREASES damage because the building moves WITH the shaking instead of fighting it.

Key Discoveries

- Stronger earthquakes cause more damage, but building design can dramatically reduce it
- Stiff buildings crack and break during earthquakes because they fight against the shaking
- Flexible buildings sway WITH the earthquake, absorbing the energy instead of breaking
- Engineers cannot stop earthquakes, but they CAN design buildings that survive them

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Earthquakes destroy all buildings equally"	Building design is the biggest factor in whether a structure survives an earthquake. In the same earthquake, a poorly designed building might collapse while a well-designed one next door suffers no damage. Modern earthquake engineering can make buildings that survive even the strongest earthquakes. Japan, Chile, and California have proven this with decades of building code improvements.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The strongest building is the stiffest building"	This is the opposite of what earthquake engineering teaches. Stiff buildings are actually MORE dangerous in earthquakes because they resist the shaking until they break catastrophically. Flexible buildings that sway with the earthquake absorb the energy gradually and survive. Think of a palm tree vs an oak in a hurricane — the palm bends and survives, the oak can snap.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We cannot do anything about earthquakes"	While we cannot prevent earthquakes, we have enormous power to reduce their impact. Better building codes, earthquake-resistant design, early warning systems, emergency preparedness plans, and community education have saved millions of lives. Countries with strong building codes (Japan, Chile, New Zealand) experience far fewer earthquake deaths than countries without them, even from stronger earthquakes.